Project Phase IV Report

On

Flex Lens

Submitted for the requirement of

Project course

BACHELOR OF ENGINEERING

COMPUTER SCIENCE & ENGINEERING

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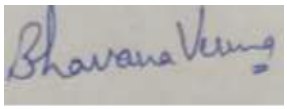

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CHANDIGARH UNIVERSITY, GHARUAN
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ABSTRACT

Object detection is one of the most basic and central tasks in computer vision. Its task is to find all the interested objects in the image, and determine the category and location of the objects. Object detection is widely used and has strong practical value and research prospects. Applications include face detection, pedestrian detection and vehicle detection. In recent years, with the development of convolutional neural network, significant breakthroughs have been made in object detection. This paper describes in detail the classification of object detection algorithms based on deep learning. The algorithms are mainly divided into one-stage object detection algorithm and two-stage object algorithm, and the general data sets and performance indicators of object indicators.

Handwritten character recognition (HCR) is the detection of characters from images, documents and other sources and changes them in machine-readable shape for further processing. The accurate recognition of intricate-shaped compound handwritten characters is still a great challenge. Recent advances in convolutional neural network (CNN) have made great progress in HCR by learning discriminatory characteristics from large amounts of raw data. In this paper, CNN is implemented to recognize the characters from a test dataset. The main focus of this work is to investigate CNN capability to recognize the characters from the image dataset and the accuracy of recognition with training and testing. CNN recognizes the characters by considering the forms and contrasting the features that differentiate among characters. Our CNN implementation is experimented with the dataset NIST to obtain the accuracy of handwritten characters

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USE OF MODERN TOOLS IN DESIGN AND ANALYSIS

Computer Vision is the branch of the science of computers and software systems which can recognize as well as understand images and scenes. Computer Vision is consisting of various aspects such as image recognition, object detection, image generation, image super-resolution and many more. Object detection is widely used for face detection, vehicle detection, pedestrian counting, web images, security systems and self-driving cars.

Handwriting digits and character recognitions have become increasingly important in today's digitized world due to their practical applications in various day to day activities. It can be proven by the fact that in recent years, different recognition systems have been developed or proposed to be used in different fields where high classification efficiency is needed. Systems that are used to recognize Handwriting letters, characters, and digits help people to solve more complex tasks that otherwise would be time-consuming and costly.

Software and Libraries Requirement:

- Python 3.7.2
- Anaconda navigator
- Jupyter Notebook
- Tensorflow
- Matplotlib
- Numpy

1. Anaconda navigator:

   Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda® distribution that allows you to launch applications and easily manage conda packages, environments, and channels without using command-line commands. Navigator can search for packages on Anaconda.org or in a local Anaconda Repository.

2. Jupyter Notebook:

   The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

3. Tensorflow:

   TensorFlow is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks.

4. Matplotlib:

   Matplotlib is a Python programming language plotting library and its NumPy numerical mathextension. It provides an object-oriented API to use general-purpose GUI toolkits suchasTkinter,wxPython, Qt, or GTK+ to embed plots into applications.

   pip install matplotlib – command

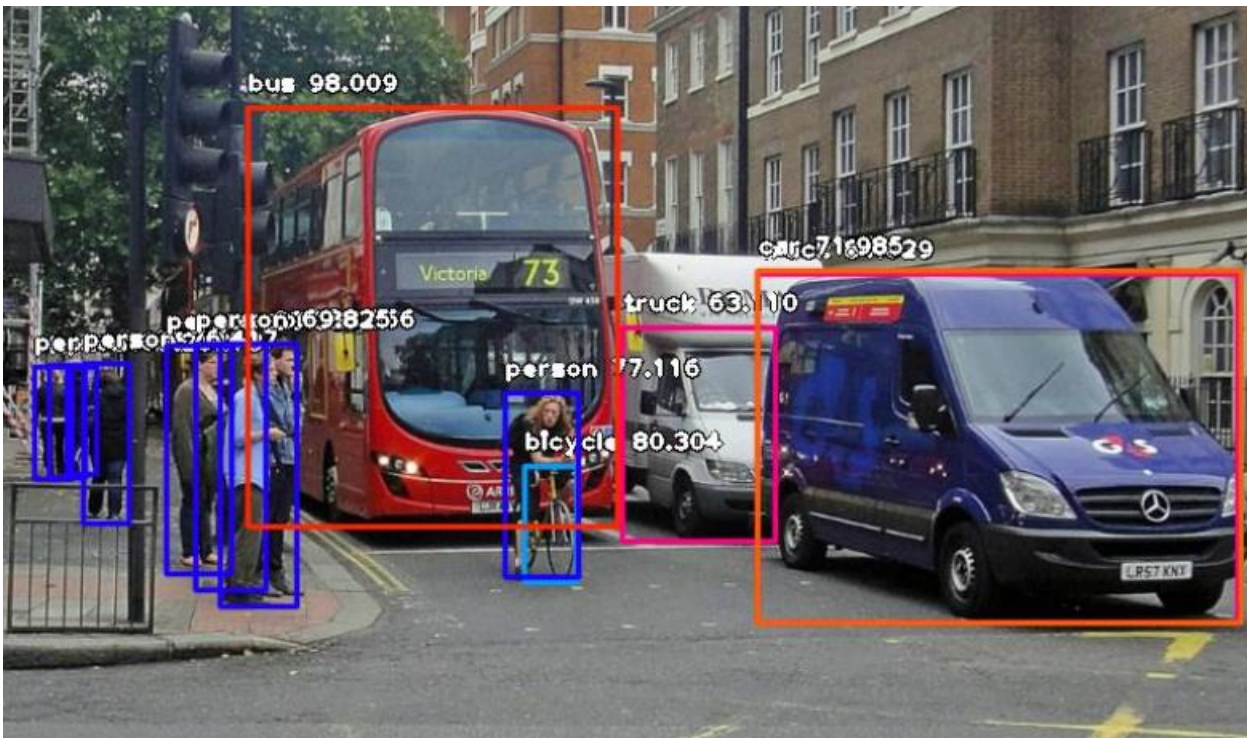
5. Numpy:

   NumPy is library of Python programming language, adding support for large, multi-dimensional array and matrices, along with large collection of high-level mathematical function to operate over these arrays. The ancestor of NumPy, Numeric, was originally created by Jim Hugunin with contributions from several developers. In 2005 Travis Olphant created NumPy by incorporating features of computing Num array into Numeric, with extension modifications. NumPy is open-source software and has many contributors.

   pip install numpy -command

Before Detection

This is a sample image we feed to the algorithm and expect our algorithm to detect and identify objects in the image and label them according to the class assigned to it



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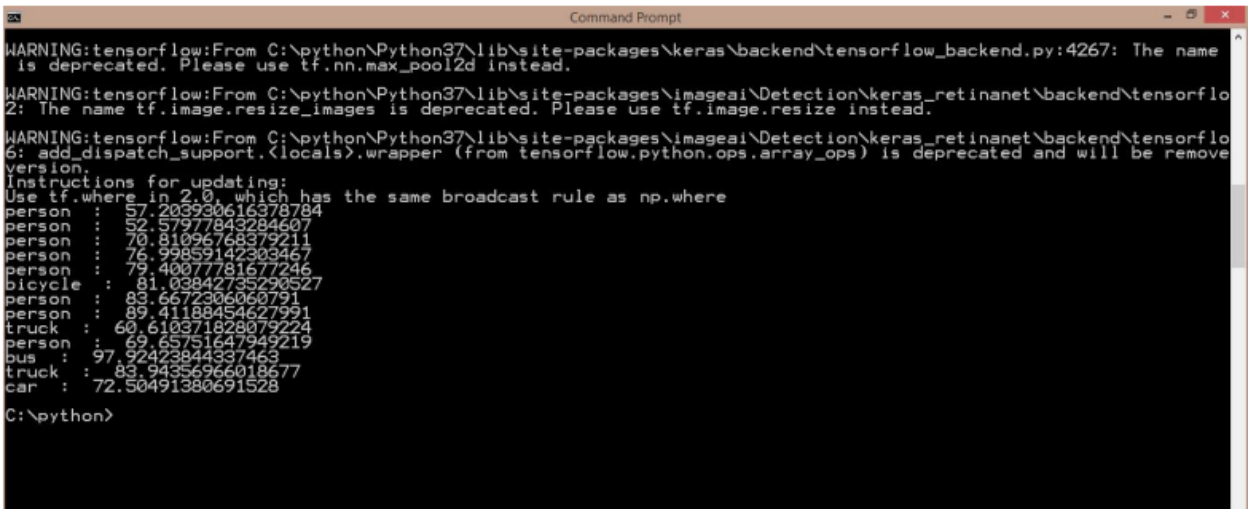
After Detection

As expected, our algorithm identifies the objects by its classes and assigns each object by its tag and has dimensions on detected image.



Console result for above image

ImageAI provides many more features useful for customization and production capable deployments for object detection tasks. Some of the features supported are:

-Adjusting Minimum Probability: By default, objects detected with a probability percentage of less than 50 will not be shown or reported. You can increase this value for high certainty cases or reduce the value for cases where all possible objects are needed to be detected

-Custom Objects Detection: Using a provided Custom Object class, you can tell the detection class to report detections on one or a few numbers of unique objects

- Detection Speeds: You can reduce the time it takes to detect an image by setting the speed of detection speed to “fast”, “faster” and “fastest”

-Input Types: You can specify and parse in file path to an image, Numpy array or file stream of an image as the input image.

-Output Types: You can specify that the detect Objects from Image function should return the image in the form of a file or Numpy array

DISCUSSION OR REPORT/RESULT ANALYSIS

Specifically, the contributions towards this project work area that outweighs existing system are as follows:

- An object recognition system is developed, that recognizes the two-dimensional and three-dimensional objects
- The feature extracted is sufficient for recognizing the object and marking the location of the object. x the proposed classifier is able to recognize the object in less computational cost
- The proposed global feature extraction requires less time, compared to the traditional feature extraction method
- The performance of the SVM-kNN is greater and promising when compared with the BPN and SVM
- The performance of the One-against-One classifier is efficient.
- Global feature extracted from the local parts of the image
- Local feature PCA-SIFT is computed from the blobs detected by the Hessian-Laplace detector.
- Along with the local features, the width and height of the object computed through projection method is used

PROJECT MANAGEMENT AND PROFESSIONAL COMMUNICATION

SLIDE 1

SLIDE 2

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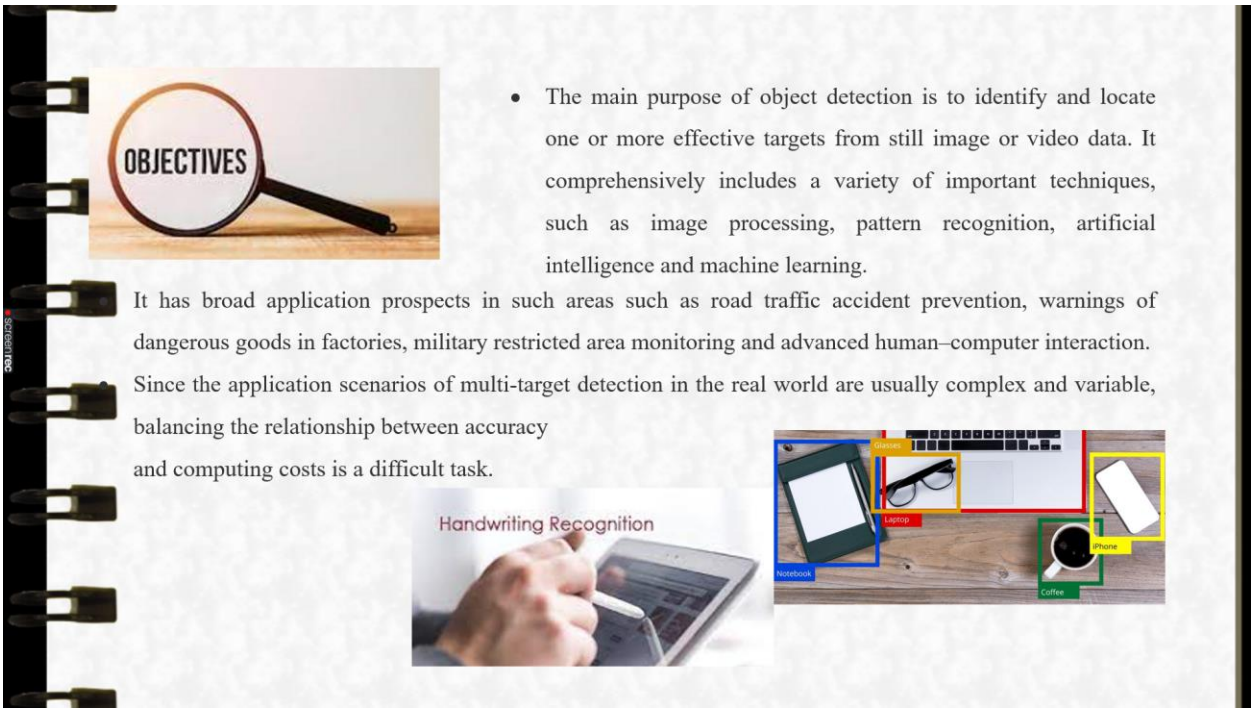

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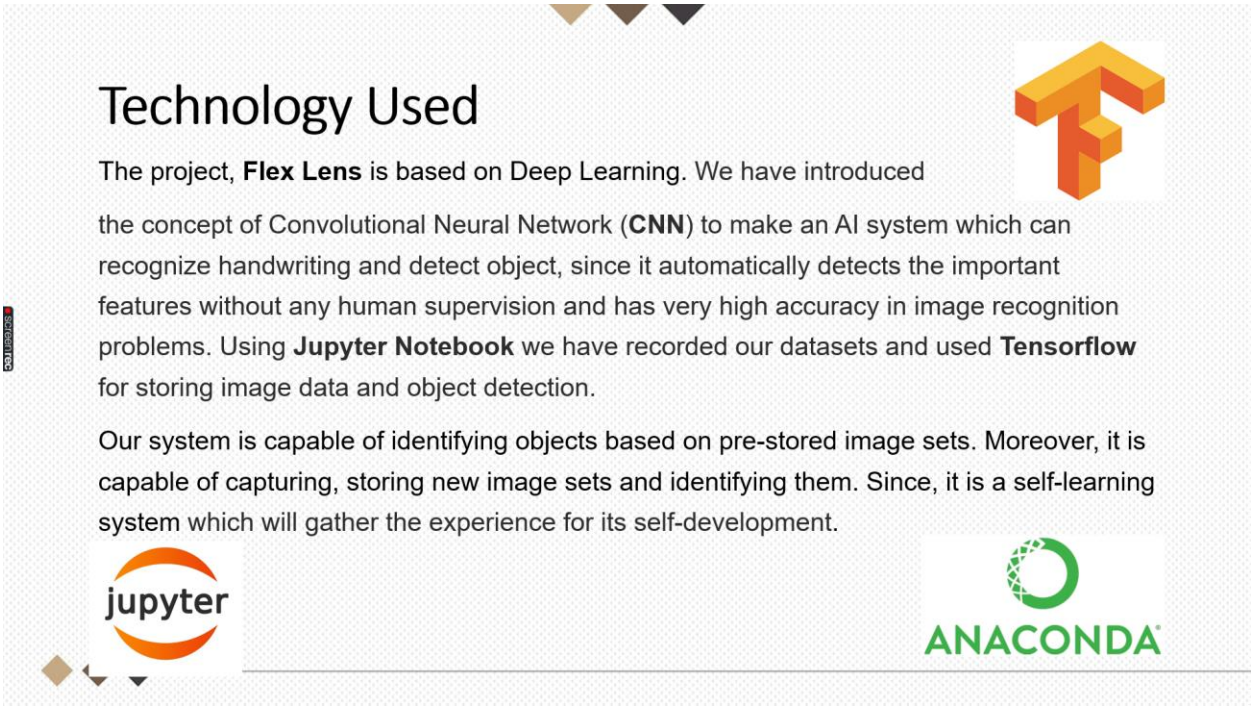

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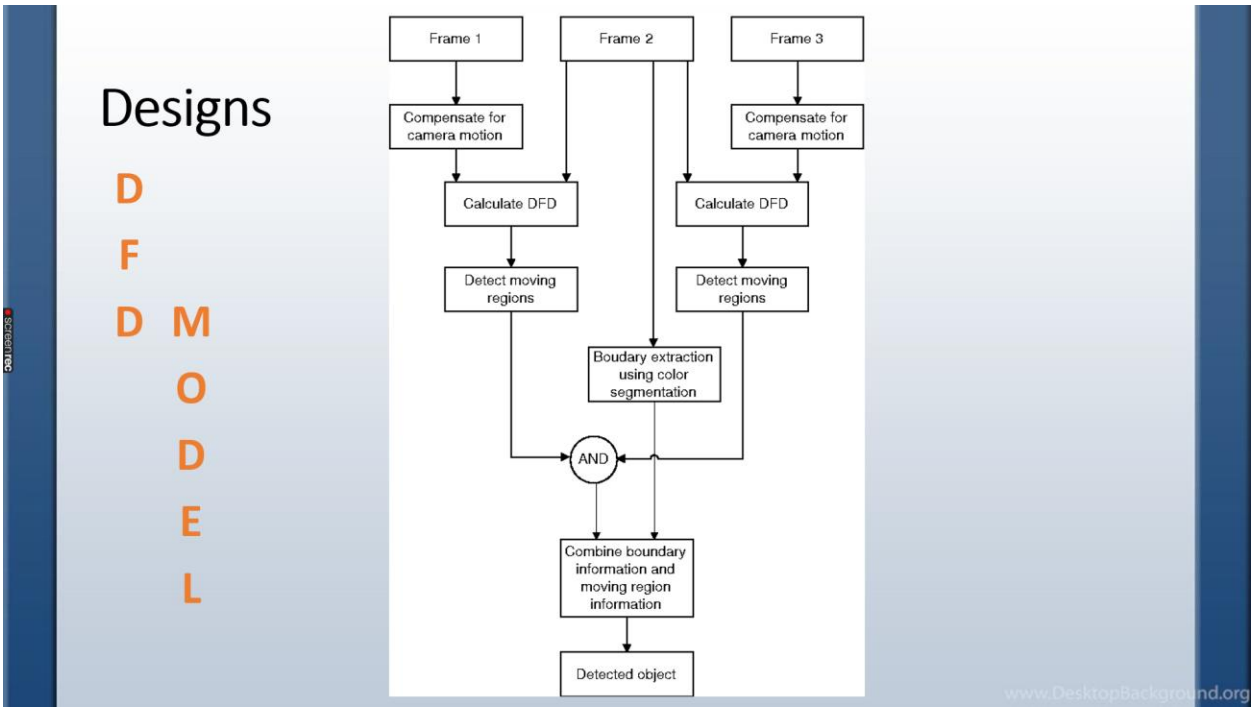

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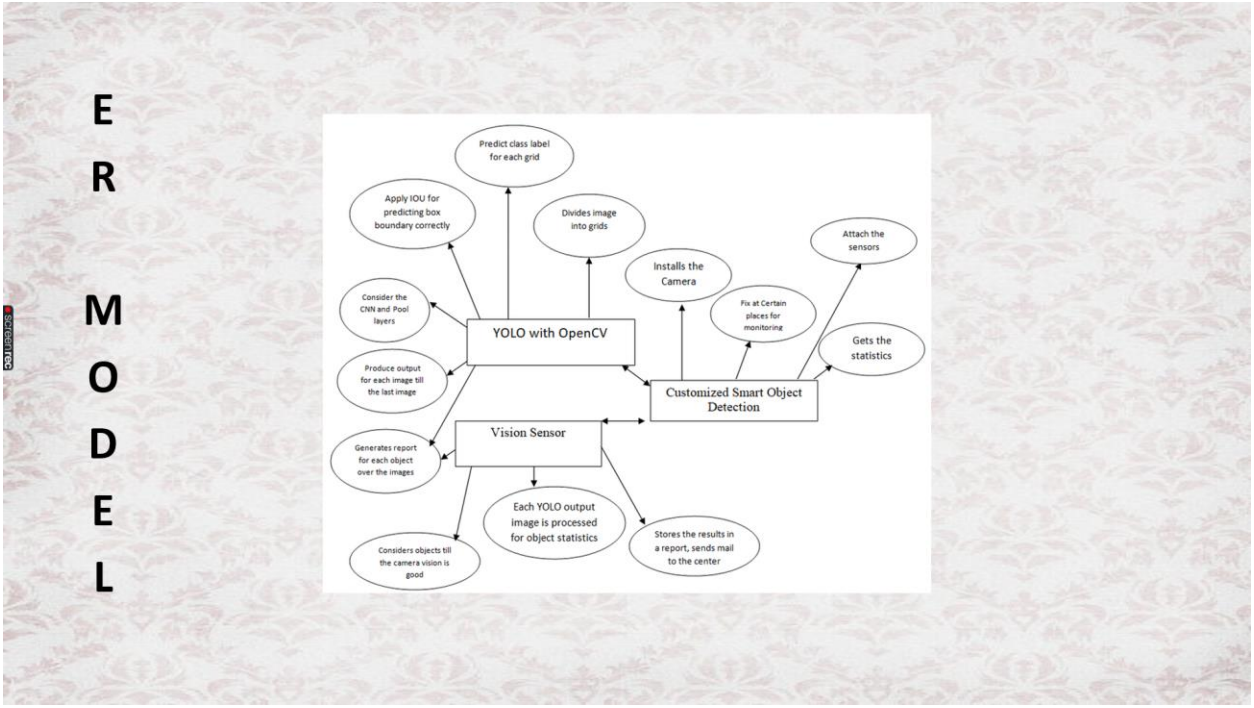

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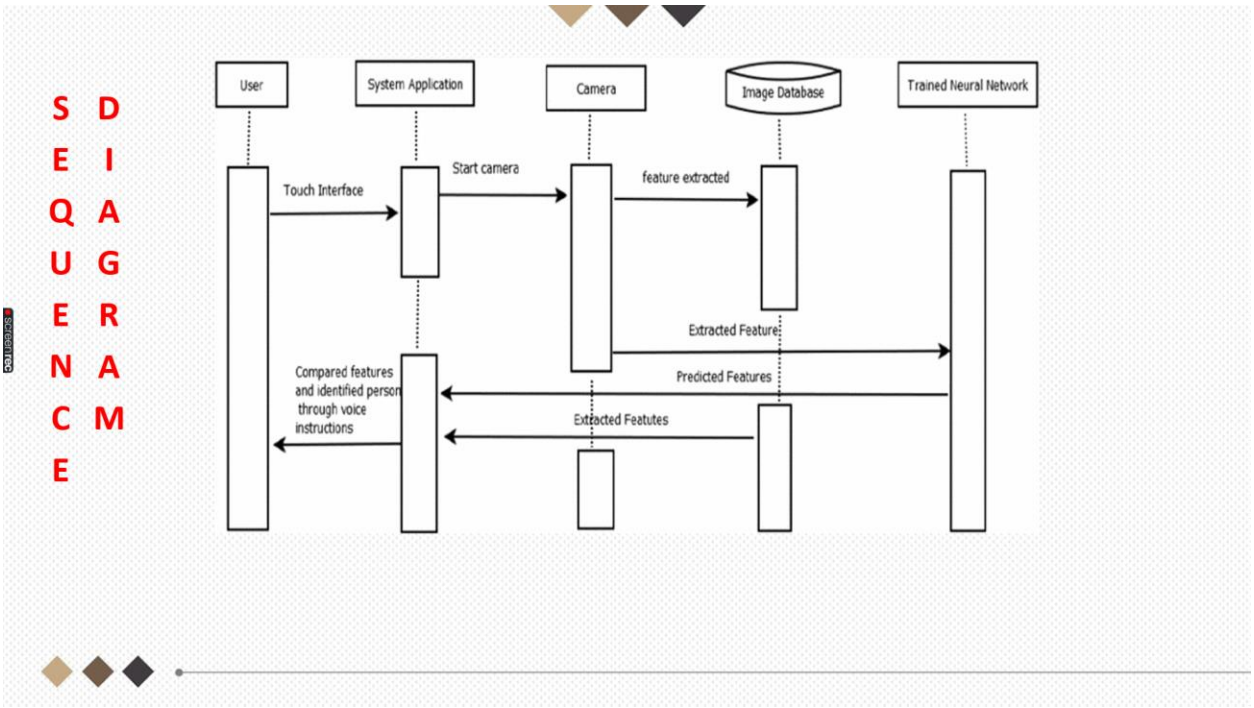

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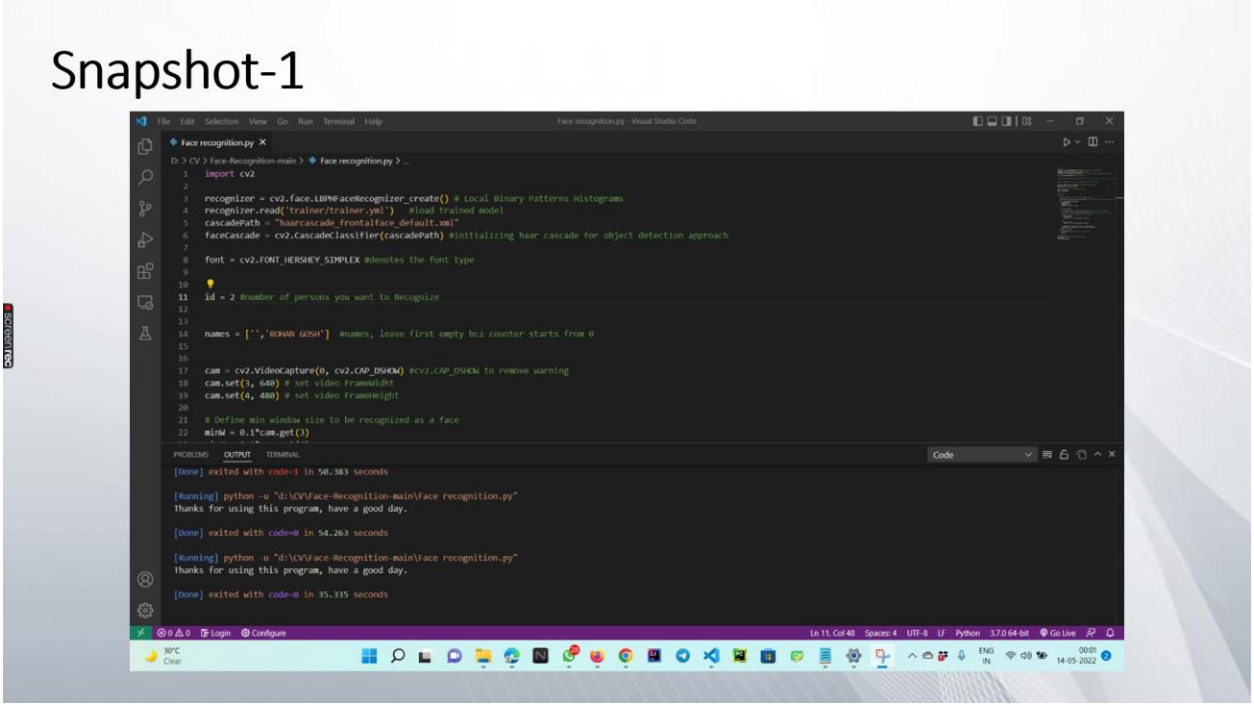

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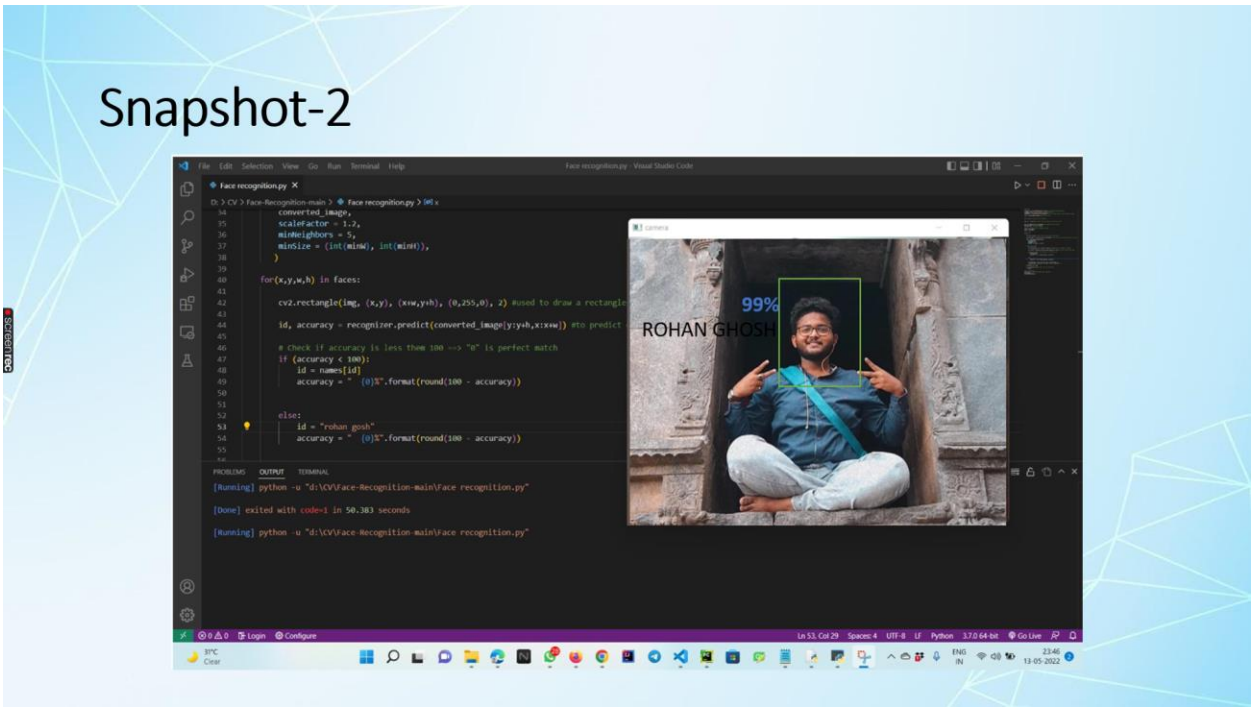

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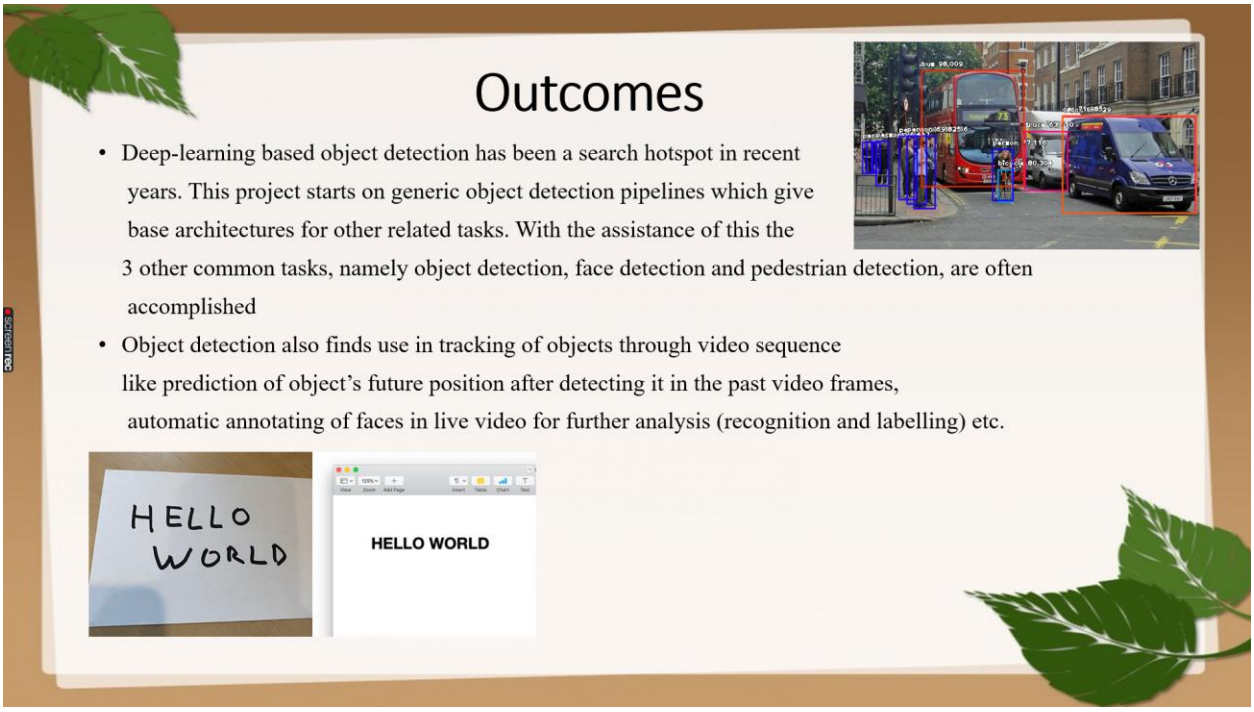

SLIDE 10

ATTAINMENT OF STATED OUTCOMES

By using this thesis and based on experimental results we are able to detect objects more precisely and identify the objects individually with exact location of an objects in the picture in x, y axis. This paper also provides experimental results on different methods for object detection and identification and compares each method for their efficiencies

Deep-learning based object detection has been a search hotspot in recent years. This project starts on generic object detection pipelines which give base architectures for other related tasks. With the assistance of this the 3 other common tasks, namely object detection, face detection and pedestrian detection, are often accomplished. Authors accomplished this by combing 2 things: Object detection with deep learning and OpenCV and Efficient, threaded video streams with OpenCV. The camera sensor noise and lightening condition can change the result because it can create problem in recognizing the objects. generally, this whole process requires GPU's rather than CPU's. But we've done using CPU's and executes in much less time, making it efficient. Object Detection algorithms act as a mixture of both image classification and object localization. It takes the given image as input and produces the output having the bounding boxes adequate to the number of objects present within the image with the category label attached to every bounding box at the highest. It projects the scenario of the bounding box up the shape of position, height and width

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